Task 1: Fine-Grained Image Classification

Data Processing & Methodology -

For this task, I utilized the Caltech-UCSD Birds-200-2011 (CUB-200) dataset. Adhering to the coursework constraints, I strictly used the provided training set for model development, creating a **90/10 split** for training and validation to tune hyperparameters. Images were resized to **300x300** pixels. To improve generalization and prevent overfitting, I implemented strong data augmentation during training,
including RandomResizedCrop, RandomHorizontalFlip, ColorJitter, and RandomRotation.

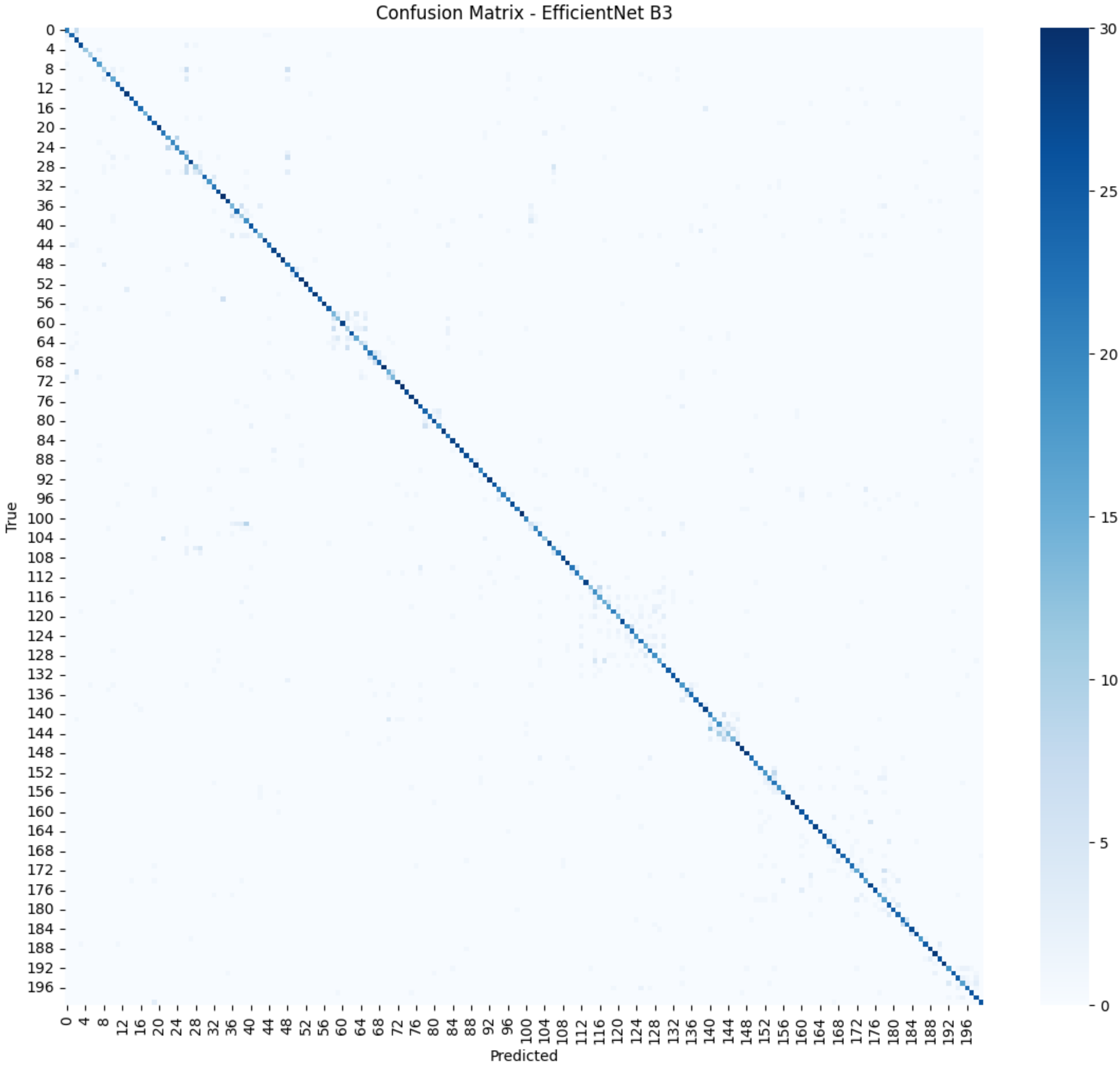
I implemented and compared two distinct approaches:

1. **Transfer Learning (EfficientNet-B3):** I utilized a pre-trained EfficientNet-B3 model (weights from ImageNet). I replaced the final classification head to match the 200 bird classes. Crucially, I employed **differential learning rates**: a higher rate (3e-4) for the new classifier and a much lower rate (1e-5) for the pre-trained backbone. This allowed the model to adapt to the specific bird features without destroying the robust pre-learned weights.
2. **Custom CNN (From Scratch):** I designed a standard 4-block Convolutional Neural Network with increasing channel depth (64 to 512). I included Batch Normalization and Dropout (p=0.5) to stabilize training.

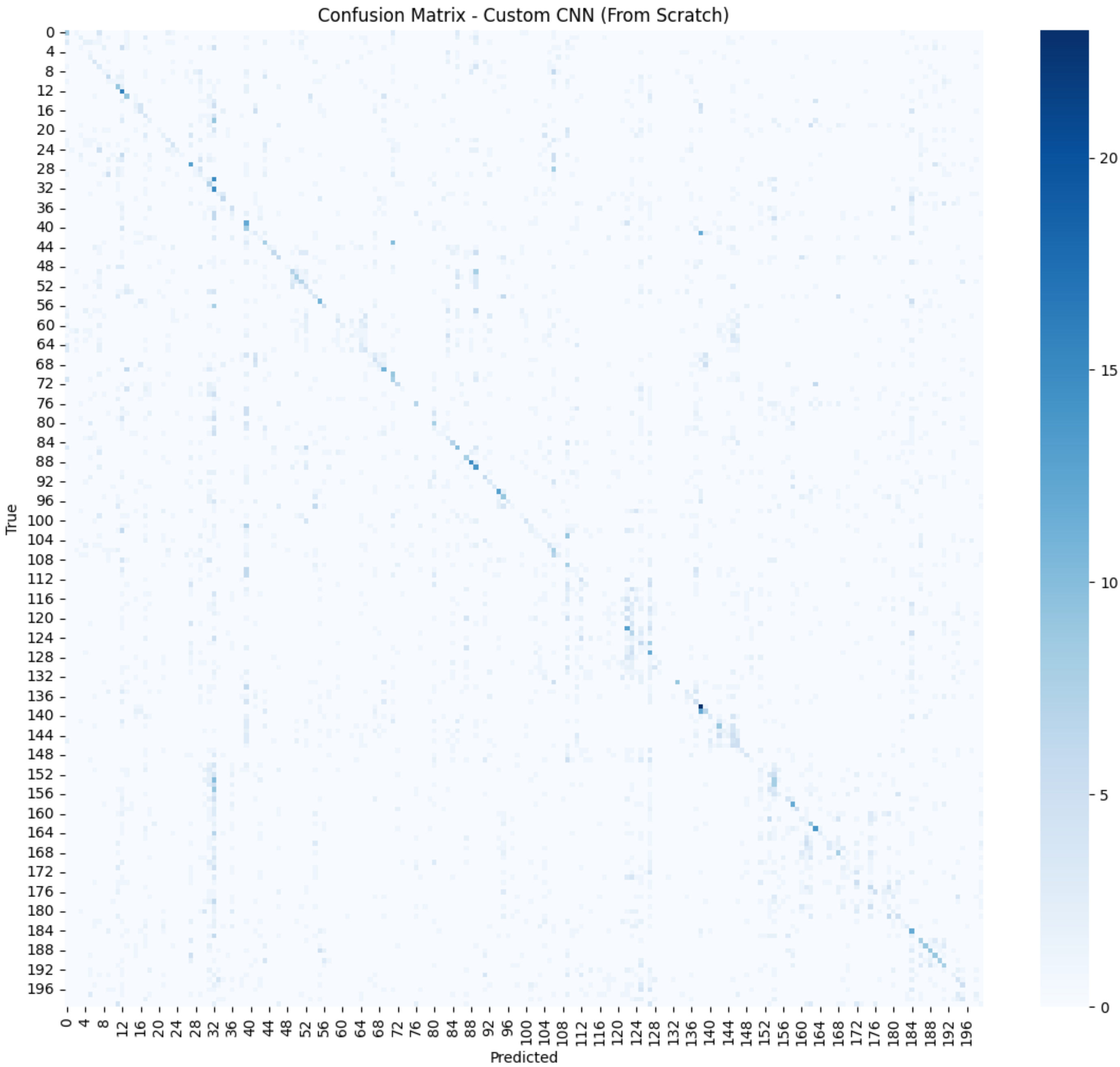
Results & Critical Analysis-

The EfficientNet-B3 model achieved a final Test Accuracy of **77.8%**, with a weighted F1-score of **0.78**. The Confusion Matrix (Figure 1) shows strong diagonal performance, though some confusion remains between visually similar species (e.g., different types of Sparrows).

In contrast, the Custom CNN achieved only **~13.2% accuracy**. This significant performance gap highlights the challenge of "Data Hunger" in Deep Learning. The CUB-200 dataset is relatively small; without the massive prior knowledge embedded in the pre-trained ImageNet weights, the custom model struggled to learn complex feature extractors (like beak shapes or feather patterns) from scratch. This validates the importance of Transfer Learning for fine-grained classification tasks with limited data.



Confusion Matrix for EfficientNet-B3 model



<u>Confusion Matrix for Custom model</u>